

Adam Mrotek

Software Engineer

✉ adam.mrotek@gmail.com 📞 +44 7517 624501  LinkedIn  GitHub  Portfolio  Personal Project

TECHNOLOGIES

TypeScript, React, Next.js, Vue 3, Node.js, Python, FastAPI, PostgreSQL, Tailwind CSS, AWS, Docker

AT WORK I'M BEST AT...

Bridging design and engineering to ship polished, type-safe interfaces fast

| Featured Project

VillageOS  www.villageos.co.uk

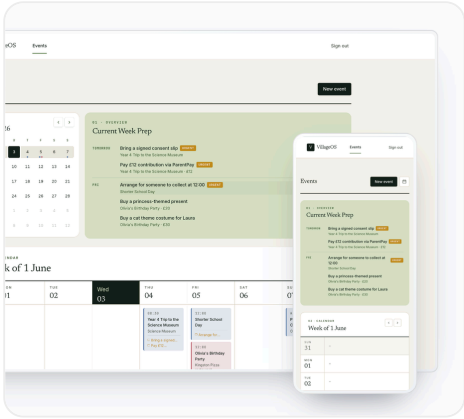
Apr 2026 – Present

Full-stack solo greenfield product

[Next.js](#) [React](#) [TypeScript](#) [FastAPI](#) [AWS Lambda](#) [Supabase](#) [OpenAI](#)

An AI-powered family OS that turns scattered parent communication — WhatsApp threads, PDF newsletters, photos of flyers — into schema-valid calendar events with action items. Built and deployed solo: frontend, backend, data, infrastructure, CI/CD and LLM evals.

- **Typed end-to-end** — one shared Pydantic v2 ↔ TypeScript contract enforces both LLM extraction and API validation, eliminating schema-drift between backend and UI.
- **Structured AI, not vibes** — instructor + Pydantic force a strict schema on every extraction, with a golden-dataset eval harness tracking per-field accuracy, latency and cost.
- **Real infrastructure** — FastAPI on AWS Lambda (arm64) via SAM, Supabase JWT auth tied to Postgres Row Level Security, deployed through OIDC GitHub Actions.



| Work experience

 **Software Engineer, Croud**

Jul 2023 – Mar 2026 (2y 8m)

Marketing Agency · Marketing

[Vue 3](#) [TypeScript](#) [FastAPI](#) [Nuxt](#) [Next.js](#) [Tailwind CSS](#) [Docker](#) [Claude / MCP](#)

- **Core Platform (Croud Control, 3,000+ users):** Owned front-end feature delivery across the marketing suite; migrated legacy Nuxt surfaces to clean, type-safe Vue 3 / TypeScript patterns.
- **Creator Platform:** Contributed front-end features and supporting FastAPI endpoints to an influencer-marketing platform built in-house by a 5-engineer team — delivering against a multi-million-pound external build quote and matching strict Figma specs.
- **Taxonomy Tool:** Built the front-end (plus supporting backend tweaks) for an internal app aligning marketing tags across teams — projected to save ~£100k/month by removing tagging inconsistency.
- **AI Tooling:** Built the front-end applications used to present internal AI tools and agentic pipelines in client pitches, plus a custom Claude sub-agent skill that converts Figma designs directly into our codebase — driving cross-team adoption and supporting upsell opportunities.

**Customer Engineer, Hullabalook**

Apr 2022 – Jul 2023 (1y 3m)

Improving retail experience using data analytics · Retail · B2B · Lifestyle · SaaS · eCommerce

React

Vue.js

Vuex

Nuxt

TypeScript

Fabric.js

Python

GCP

- Developed and launched features and multiple apps as part of cross-functional product teams, shipping through CI/CD pipelines.
- Constructed robust applications using React, Vue, Vuex and Nuxt, plus highly visual interfaces built with Fabric.js and the HTML5 Canvas API.
- Enhanced data accuracy by investigating programmatic solutions for unstructured, incomplete datasets using Python, Google Colab and Google Cloud Platform (GCP).
- Prototyped features that resulted in successful upsells across multiple clients, and improved Lighthouse / SEO scores to enhance app performance.

**Architectural Assistant (Part I & II), Bryden Wood**

Mar 2018 – Sep 2021 (3y 6m)

Multidisciplinary architectural company · Property · Sustainability · Design · Visualisation · Construction

Adobe Creative Suite

Photoshop

InDesign

BIM

- Designed and contributed to architectural projects for airports and data centers, applying platform-based, systems-thinking design principles.
- Facilitated cross-functional collaboration between architectural and mechanical / electrical design teams to achieve project objectives.

| Education

Anthropic Academy

May 2026

Certificate of completion — Claude with Amazon Bedrock

- Building production-ready AI applications with Claude on Amazon Bedrock — the Bedrock API, prompt engineering and evaluations, tool use, retrieval-augmented generation (RAG), the Model Context Protocol (MCP), and agents.

MSc & BArch, Architecture

2011 – 2017

Politechnika Poznańska (Poznan University of Technology) · incl. Erasmus, Università di Firenze

| More about me

I'm looking to move roles because...

I'm looking for deeper ownership across the whole stack and harder engineering problems — particularly building real products on top of LLMs.

Interests

Parenthood tricks, feng-shui, cooking.

Right now I'm learning...

AI pipelines and multi-agent workflows, RAG, and app deployment.

Languages

English (Full Professional), Polish (Native / Bilingual).